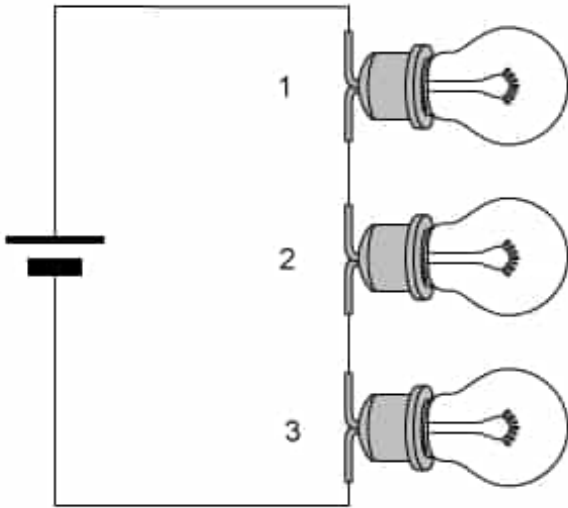




Three different light bulbs of varying brightness are connected in series to a voltage source. Bulb 1 is the brightest, and bulb 3 is the dimmest. The voltages across the bulbs, V_1 , V_2 , and V_3 , are related by which of the following?

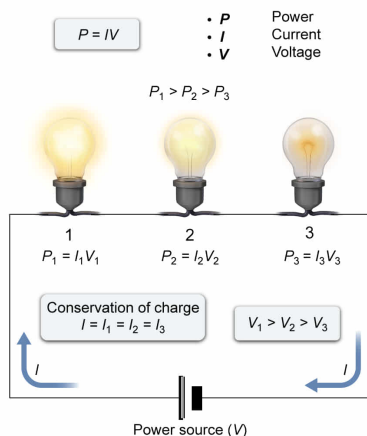
- ✓ ☒ A. $V_1 > V_2 > V_3$ [70%]
☐ B. $V_1 = V_2 = V_3$ [20%]
☐ C. $V_3 = V_1 > V_2$
☐ D. $V_3 > V_2 > V_1$ [8%]

Correct

71% answered correctly

Explanation:

Electric circuit of light bulbs with various brightness connected in series



In an electric circuit, a power source with a given voltage generates a **current**, which is the flow of charge per unit time within a conductor (eg, wire, bulb, resistor). When circuit elements are **in series**, they are connected end-to-end in a closed loop.

Conservation of charge applies to a series circuit, implying that the magnitude of current flow is the same through each element.

When current flows through a conductor, the voltage will drop across the conductor due to resistance, which causes the conductor to heat up. **Ohm's law** implies that resistance R is the ratio of the voltage drop V to the current I :

$$R = \frac{V}{I}$$

The voltage drop results in **power dissipation** in the form of heat and, in the case of a light bulb, radiated light. Power P is the rate at which energy is transferred to the conductor and is the product of I and V :

$$P = IV$$

In this question, the brighter bulbs are dissipating power at a greater rate than the dimmer bulbs. Since the power is greatest for the brightest bulb (bulb 1) and least for the dimmest bulb (bulb 3), the relative power of the bulbs is:

$$P_1 > P_2 > P_3$$

The bulbs are connected in series. Hence, the current flow is the same through each:

$$I_1 = I_2 = I_3 = I$$

Using the equation for power and substituting the corresponding I and V gives:

$$I_1 V_1 > I_2 V_2 > I_3 V_3$$

$$I V_1 > I V_2 > I V_3$$

$$V_1 > V_2 > V_3$$

Therefore, the current through each bulb is equal, the brightest bulb must have the greatest V , and the dimmest bulb must have the smallest V .

(Choices B, C, and D) The current flow through each bulb is equal. However, the voltage drop is greatest across bulb 1 and smallest across bulb 3.

Educational objective:

Current flow is equal through all elements of a series circuit. Light bulb brightness is proportional to electric power, which is the product of the current and voltage.

Time spent: 0 sec

Subject:
Physics

QID: 403907

System:
Electrostatics and Circuits

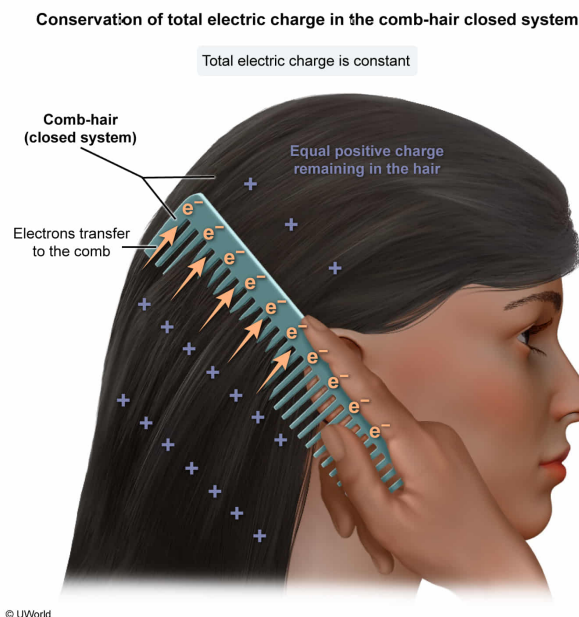
A student pulls a plastic comb through her hair on a dry day. Afterward, her hair is attracted to the comb. Which of the following occurs when the comb passes through the hair?

- ☐ A. Total charge is conserved, and no charges are transferred between the hair and the comb. [3%]
- ✓ ☒ B. The comb gains electrons and the hair becomes positively charged. [89%]
- ☐ C. The comb gains electrons and the hair becomes negatively charged. [6%]
- ☐ D. Both the comb and hair become positively charged.

Correct

88% answered correctly

Explanation:



The law of **conservation of charge** states that electric charge is neither created nor destroyed. A neutral object has an *equal* amount of **positive and negative charge**. When two objects are rubbed together, **electrons** can transfer from one object to another. Objects that gain electrons will have an excess negative charge (ie, become negatively charged), and objects that lose electrons will have an equal amount of excess positive charge (ie, become positively charged). The total charge of the system remains constant, but the distribution of charge among the objects in the system is different.

In this question, the comb and hair are initially neutral. However, after the comb passes through the hair, the hair and comb are oppositely charged because they attract each other. Therefore, negatively charged electrons are transferred from the hair to the comb, making the comb negatively charged and the hair positively charged, consistent with conservation of charge.

(Choice A) The total charge is constant, but charge is transferred between the hair and the comb, making the hair positively charged and the comb negatively charged.

(Choice C) The hair cannot become negatively charged if electrons are transferred from the hair to the comb.

(Choice D) Positive charges on both the comb and hair would repel each other. A net positive charge is also inconsistent with charge conservation.

Educational objective:

The total amount of charge in a system is constant, but charges can be transferred within the system, making some objects positively charged and others negatively charged.

Time spent:0 sec

Subject:

Physics

QID:403908

System:

Electrostatics and Circuits

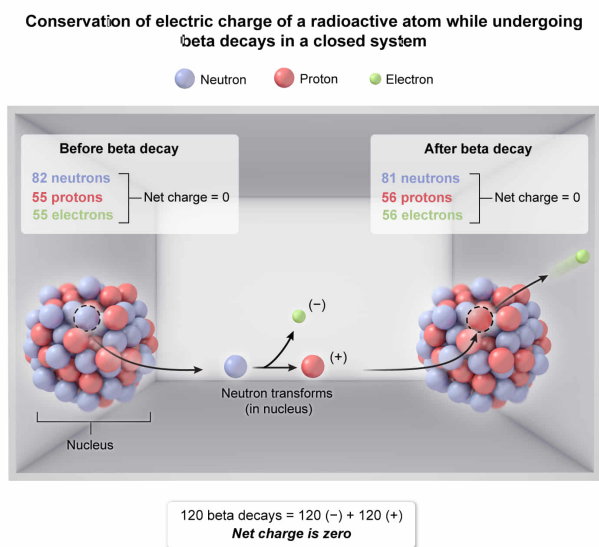
During the beta decay process, a neutron transforms into an electron and a proton, and the electron is ejected out of the atom. A closed system containing a number of atoms is electrically neutral and undergoes 120 beta decay processes. What is the resulting net charge of the closed system? (Note: the elementary charge $e = 1.6 \times 10^{-19} \text{ C}$)

- ✓ ☒ A. 0 C [36%]
☐ B. $+1.92 \times 10^{-17} \text{ C}$ [39%]
☐ C. $-1.92 \times 10^{-17} \text{ C}$ [19%]
☐ D. $+3.84 \times 10^{-17} \text{ C}$ [4%]

Correct

37% answered correctly

Explanation:



According to the law of **conservation of electric charge**, electric charge cannot be created or destroyed; the total charge is conserved in any process. Moreover, only heat—not matter—can be exchanged with the surroundings in a **closed system**. Therefore, in a closed system, the total charge must remain constant.

In this question, a radioactive nucleus undergoes 120 **beta decay** processes, where a neutral neutron transforms into one electron and one proton. Consequently, each beta decay process produces the same positive and negative charges.

In addition, the system is **closed** and **neutral** before the beta decay processes. Due to the conservation of charge and the closed system, the charge balance is maintained. The system will always be neutral regardless of the number of beta decay processes the atom undergoes within the system. Therefore, the **net charge** of the closed system is **zero**.

(Choice B) $+1.92 \times 10^{-17} \text{ C}$ is the charge of the 120 protons produced by the 120 cycles of beta decay and not the overall charge.

(Choice C) $-1.92 \times 10^{-17} \text{ C}$ is the charge of the 120 electrons produced by the 120 cycles of beta decay and not the overall charge.

(Choice D) Mistakenly adding the magnitude of the charges of the electrons and protons results in a total added charge $+3.84 \times 10^{-17} \text{ C}$. However, it is necessary to consider the sign of each charge.

Educational objective:

Electric charge is conserved; charge cannot be created or destroyed. In a closed system, only heat—not matter—can be exchanged with the surroundings, and the total charge is constant.

Time spent:0 sec

Subject:
Physics

QID:403909

System:
Electrostatics and Circuits